

Beyond FOMO

An Agentic AI-Based Information Filtering and Opportunity Assistant

Developed Beyond FOMO, an Agentic AI information assistant that reduces information overload by filtering and prioritizing relevant updates across jobs, scholarships, visas, AI, technology, and government notifications. Built a LangChain and LangGraph-based multi-agent workflow for information retrieval, relevance analysis, content moderation, priority classification, and opportunity detection. Integrated Hugging Face models for image-to-text extraction and voice-based interaction with speech-to-text and text-to-speech capabilities. Applied prompt engineering and multilingual processing to deliver personalized, concise, and actionable information while filtering duplicate, irrelevant, clickbait, toxic, and hateful content.

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Project Report

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1 Abstract

Beyond FOMO implements this idea as an Agentic AI information assistant. The system combines LangChain and LangGraph with an LLM, search and category tools, relevance and priority processing, moderation, personalization, multilingual translation, voice input, text-to-speech, and image-to-text processing. The implemented application uses Streamlit for the interface, Gemini 2.5 Flash as the LLM, LangGraph checkpointing with SQLite for conversational state, Whisper for voice transcription, Hugging Face TrOCR for printed-text extraction from images, and Edge TTS for spoken responses. The conceptual architecture is organized around more than ten functional agentic responsibilities when graph nodes and specialist tools are considered together; the current codebase specifically implements eight callable specialist tools and five graph-level processing nodes. This distinction is stated explicitly to keep the report technically accurate.

2 Introduction

The modern world of information creates a situation. Social media makes it easy to find information. The same platforms can show users a never-ending flow of things to compare themselves to things that are exaggerated, arguments, ads, repeated updates and stuff that makes people feel emotional. A person can end up thinking they are missing out even though they are taking in much information.

The project starts with an idea: stopping using social media can get rid of one kind of FOMO but not always the worry about missing important information. The project report that comes with this work talks about two cases: FOMO while using media and FOMO after stopping social media use. In the case people might compare their regular lives with the best parts people show online. In the case the worry is about missing helpful chances and official messages. The original document clearly says the problem is about finding a way to take in the information that matters while leaving out the rest that is not needed or harmful.

This leads to a project called Beyond FOMO, an AI-based information helper that tries to find the parts, from the unimportant parts. Of making a person go through many feeds the system uses special parts to search, filter, summarize, rank, control, translate and explain the information.

3 Research Background and Definition of FOMO

The academic literature commonly defines FOMO as a pervasive apprehension that others may be having rewarding experiences from which an individual is absent. A foundational paper by Przybylski et al. introduced a widely used conceptualization and measurement approach for FOMO in the context of social experience and digital engagement.

In this project, however, the term FOMO is used in a broader engineering sense. The project distinguishes between:

1. **Social FOMO:** concern that other people are experiencing rewarding social activities, achievements, or lifestyles without the user.
2. **Information FOMO:** concern that the user may miss information, opportunities, rules, deadlines, or developments that could materially matter to them.

The second category should be understood as a **project-defined operational concept**, not as a formally established clinical subtype of FOMO. The purpose of this distinction is important: the system is an information-management and digital-well-being solution, not a medical treatment or diagnostic system.

4 Problem Scenario: Case 1 – FOMO While Using Social Media

Consider a user opening Instagram during a weekend and seeing friends posting photographs from a group bike ride, trip, party, or vacation. The user may interpret the post as evidence that an important experience has happened without them. Similar triggers can occur on professional and other platforms.

Examples include:

- LinkedIn posts announcing a new job, admission to a top university, promotion, award, or professional achievement.
- Facebook posts showing office parties, trips, celebrations, or social gatherings.
- Snapchat streaks and frequent lifestyle updates.
- Highly polished daily-life content from YouTubers or influencers that encourages unrealistic comparison.

- Arguments, hateful speech, toxic discussions, and politically charged comment sections that repeatedly expose users to negative content.

The supplied project research notes that repeated exposure can be associated with stress, anxiety, reduced self-esteem, sleep disturbance, lower productivity, and constant comparison. These should be presented as potential consequences reported or discussed in the project’s research framing, not as deterministic outcomes for every user.

A key mechanism is **asymmetric comparison**: a user often compares their complete, ordinary day with selected highlights from other people’s lives. The result is a distorted comparison between private reality and public presentation.

4.1 Typical Triggering Language

The system should be particularly careful with language that creates urgency, exclusion, scarcity, or social comparison. Examples include:

- “Everyone is there”
- “You are missing out”
- “Everyone is getting this”
- “Last chance”
- “Only today”
- “Limited seats”
- “Breaking”
- “Urgent”
- “Exclusive”
- “Don’t miss this”
- “Everyone already knows”
- “Apply now”

These words are not inherently harmful. For example, a genuine scholarship deadline may legitimately require urgency. The engineering objective is therefore not to delete every urgency word, but to distinguish **actionable urgency** from **engagement-driven urgency**. This distinction is central to the project’s anti-noise philosophy.

5 Problem Scenario: Case 2 – FOMO After Quitting Social Media

After recognizing the negative effects of excessive social-media consumption, a user may quit or substantially reduce social-media use. The project research describes an initial improvement: less distraction, greater calm, and more focus on studies, work, and personal life.

However, after some time, a different concern can emerge: *What if I am missing something important?*

Typical questions include:

- What if an important visa policy changes?
- What if a university announces a scholarship?
- What if immigration rules change?
- What if an internship opportunity appears and closes quickly?
- What if an examination deadline is announced?
- What if a company announces a hiring drive?
- What if an important AI or technology development occurs?

This is the central design insight of Beyond FOMO: **information avoidance is not necessarily the solution to information overload**. The user needs a controlled information layer that preserves access to valuable signals without recreating the infinite-scroll environment.

6 Problem Statement

Using social media and completely avoiding social media both have advantages and disadvantages. Social platforms can provide access to useful information and opportunities, but they can also create distraction, negative exposure, misinformation, repetitive content, and unhealthy comparison. Quitting social platforms can improve focus and reduce exposure to noise, but it can also increase uncertainty about whether important updates or opportunities are being missed.

Therefore, the engineering problem is:

How can an AI system provide a user with only the information that is relevant, timely, trustworthy, actionable, and personalized, while minimizing unnecessary social comparison, toxic content, clickbait, repetition, and information overload?

The proposed answer is an Agentic AI information assistant that acts as a controlled information gateway between the user and a large, continuously changing information environment.

7 Proposed Solution: Beyond FOMO

Beyond FOMO is designed as a multi-component Agentic AI system. Instead of exposing the user to hundreds of posts or search results, it attempts to perform the intermediate work of information acquisition and reduction.

The conceptual pipeline is:

User Need → Search/Discovery → Filtering → Relevance → Priority → Moderation → Summarization

Optional interaction layers then provide translation, voice input, text-to-speech, and image-to-text extraction.

The system is intended to support sectors such as:

- AI and Machine Learning
- Jobs and internships
- Study abroad
- Scholarships
- Visa and immigration updates
- Government notifications
- Technology news
- Research papers
- Startup and business news
- Weather and other quick utilities

8 Agentic Architecture

The original project design identifies specialist responsibilities including source discovery, content filtering, relevance analysis, FOMO priority scoring, conversation, translation, voice processing, and opportunity detection. The implementation expands this into a graph-based workflow with routing, tool execution, moderation, preference handling, and conversational response generation.

8.1 Functional Agentic Responsibilities

A research presentation may describe the system as having more than ten functional agentic responsibilities when specialist tools and graph-level agents are counted separately. The major responsibilities are:

Component	Responsibility
Source Discovery	Retrieve current information from search/news sources.
Category Retrieval	Focus searches on categories such as AI, jobs, visas, scholarships, and business.
Web Research	Handle evergreen facts, definitions, comparisons, rankings, and how-to questions.
Content Filtering	Remove or reduce duplicate, irrelevant, clickbait, spam, gossip, and toxic material.
Relevance Processing	Align retrieved information with user preferences and requested topics.
FOMO Priority	Classify information as Critical, Important, Can Wait, or Ignore.
Opportunity Detection	Extract scholarships, jobs, internships, visa updates, deadlines, and government announcements.
Summarization	Convert retrieved information into concise, readable points.
Moderation	Clean tool outputs by removing hate speech, toxic material, unverified claims, source clutter, and promotional noise.
Intent Routing	Distinguish a query from feedback about future response preferences.
Preference Management	Maintain category, style, and language preferences.
Conversation	Produce natural responses and follow-up explanations.
Translation	Translate responses into supported target languages.
Voice Input	Convert recorded speech into text.

Text-to-Speech	Convert the latest AI response into spoken audio.
Image-to-Text	Extract printed text from an uploaded image.
Weather Utility	Retrieve current weather information for a requested city.

8.2 Implementation Accuracy Note

The current supplied backend code contains **eight callable specialist tools**: I am looking at a list of things that a project can do. It has a SearchEngine, a WebSearch, a WeatherTool, a TranslateTool, a CategoryNews, a SummarizeTool, a FOMOPriority and an OpportunityFinder. The project also has some important parts like IntentRouter, ChatAgent, ModerateNode, FeedbackNode and ToolNode. So it is better to say that the project has than 10 things that can do stuff rather than saying it has more than 10 independent things that can work on their own. This way we are being honest about what the project can do, without making it sound like it can do more than it really can. The project is still made to work like it has many agents working together.

9 Core Agent Responsibilities

9.1 Source Discovery and Search

The system looks for the news and information. It checks for news, events, deadlines and announcements. The system has a search tool and a separate way to search the web.

The system tells the model to use these tools to find information that's time sensitive, such as news, jobs, scholarships, visas government updates, weather and markets. The system uses these tools to find the information, about news, jobs, scholarships, visas government updates, weather and markets.

9.2 Content Filtering and Anti-Noise Layer

The moderation stage is made to get rid of things like hate speech and gossip. It also removes things that're not true and names of sources that are not needed. Things like "Read are also taken out because they are not important. The goal of the moderation stage is to keep the information and remove all the extra stuff that is not useful. This is a part of the project, which is to filter out the bad information so people can still get the information they need. The moderation stage is really, about filtering the noise not about stopping people from getting information.

9.3 Relevance and Personalization

The system keeps track of what you like including the types of things you want to see how you want things explained and the language you want to use. So when you give feedback like asking for points or saying you want to see things about jobs or artificial intelligence or visas or scholarships it can change what you see next. The system uses your feedback like your preference, for jobs or artificial intelligence or visas or scholarships to make the things you see more useful.

9.4 FOMO Priority Agent

The priority layer is the heart of the project. Instead of presenting a long list, information is classified into four levels:

1. **Critical** – information that may require prompt attention.
2. **Important** – useful information that should not be ignored.
3. **Can Wait** – relevant but not urgent information.
4. **Ignore** – low-value information for the current user.

This is the key transformation from a conventional news reader to a decision-support assistant.

9.5 Opportunity Detection

The Opportunity Finder is specifically designed to find scholarships, jobs, internships, visa updates, deadlines and government announcements. The project therefore focuses on information rather than news consumption, for its own sake.

10 LangChain and LangGraph Integration

LangChain is what we use to make our tools and messages work together with the language model. On the backend we use a Gemini 2.5 Flash chat model. It helps us connect our specialist tools to the language model.

We use LangGraph to show how our workflow works. It is like a map that shows us where we are and where we need to go. This map starts with something called IntentRouter.

When someone asks for feedback it goes to FeedbackNode. When someone asks for information it goes to ChatNode. If ChatNode needs to use a tool it moves to ToolNode to ModerateNode and finally back, to ChatNode so we can get a response. We use something called SQLite checkpoint to save our place in the conversation so we can pick up where we left off when we use Streamlit again.

Conceptually:

$$\text{START} \rightarrow \text{Intent Router} \rightarrow \begin{cases} \text{Feedback Node} \rightarrow \text{END} \\ \text{Chat Agent} \rightarrow \text{Tool Node} \rightarrow \text{Moderation} \rightarrow \text{Chat Agent} \end{cases}$$

This graph-based design is more appropriate than a single prompt because different responsibilities can be isolated, inspected, and modified independently.

11 Prompt Engineering

Prompt engineering is used to establish behavioral rules for the FOMO Guardian. Important instructions include:

- use tools for current or time-sensitive information;
- distinguish current information from evergreen questions;
- produce concise, useful bullet points;
- avoid inventing current facts when tools are unavailable;
- remove source names and unnecessary source clutter from final responses;
- ask for preferences before setting up daily insights;
- maintain conversational continuity.

The moderation prompt is particularly important because it creates a second-stage safety and quality layer after tool retrieval.

12 Multilingual and Audio Processing

The project has a part that helps people who speak different languages talk to each other. This part can translate things into English, Telugu, Hindi, French and Italian. So if you get some information in one language you can see it in the language you like better.

The application also does something with sound. You can talk to it. It will write down what you say. Then it puts what you said into the message box so you do not have to type it all out. This makes it feel more natural to ask the application things because you can just talk to it of having to type everything out.

Implementation qualification: the supplied application currently loads `openai/whisper-small.en`, so the implemented speech-to-text model is English-specific. Therefore, the report should describe the project as having a multilingual translation/output capability and an audio interaction pipeline, while avoiding the technically unsupported claim that the current Whisper configuration performs unrestricted multilingual speech recognition.

13 Text-to-Speech Output

The application includes a SPEAK function that reads the latest AI response aloud. Edge TTS is used to generate an MP3 response, which is then presented through the Streamlit interface. This creates a complete interaction loop:

Voice Query → Speech-to-Text → Agentic Processing → AI Response → Text-to-Speech.

The current implementation uses the `en-US-GuyNeural` voice. A future version can map the selected output language to an appropriate neural voice.

14 Image-to-Text Capability

The application also does image-to-text generation. You can upload a picture that's a PNG, JPG, JPEG or WEBP image. The system uses the Hugging Face microsoft/trocr-printed model to look at the image and find the printed text. It puts the text it finds into the chat box so you can look at it before you send it to FOMO Guardian.

This feature means the system can do more than answer text and voice questions. For example you can take a picture of a notice or a deadline or something that is printed and ask the assistant to tell you what it means or give you a version of the image-to-text generation result from the image-, to-text generation.

15 User Interface and Application Layer

The frontend is implemented using Streamlit. The interface includes:

- Beyond FOMO branding and a FOMO Guardian assistant identity;
- category-oriented feature cards;
- conversational message bubbles;
- text input;
- image attachment;
- microphone-based voice input;
- SPEAK output;
- Submit control;
- responsive layout and visual status indicators.

The supplied application also contains a dedicated visual “Guardian” mascot and a glass-style interface designed to communicate the concept of a calm information gateway rather than an infinite social feed.

16 End-to-End Workflow

A typical user interaction proceeds as follows:

1. The user asks a question by typing, speaking, or uploading an image.
2. If voice is used, the audio is converted to text.
3. If an image is uploaded, printed text is extracted using TrOCR.
4. The query enters the LangGraph workflow.
5. IntentRouter determines whether the message is a query or preference feedback.
6. For a query, ChatAgent interprets the request and selects a suitable tool when necessary.
7. Search, category retrieval, weather, translation, summarization, priority, or opportunity tools are invoked as appropriate.
8. Tool results pass through the moderation layer.
9. The cleaned information is returned to the conversation agent.
10. The final answer is shown in the chat interface.
11. The user can request SPEAK to hear the generated response.

17 What Makes Beyond FOMO Different?

The project's novelty is not simply “using an LLM”. Its central design principle is **information reduction with preservation of important signals**.

A normal social feed is designed to keep people watching and interacting. A normal search engine gives lots of answers. A normal chatbot can give an answer to a question. Doesn't always focus on cleaning up the information that a person sees.

Beyond FOMO instead attempts to answer a different question:

“What is the small amount of information I actually need to know?”

The FOMO Priority and Opportunity components are particularly important because they transform raw information into actionable signals.

18 Advantages

- Reduces unnecessary information overload.
- Provides personalized information instead of a generic feed.
- Separates current-information retrieval from evergreen search.
- Adds a moderation layer after external retrieval.
- Prioritizes information instead of displaying everything equally.
- Highlights actionable opportunities such as scholarships, jobs, internships, and deadlines.
- Supports multilingual translation.
- Supports voice input and spoken AI responses.
- Supports image-to-text interaction.
- Uses LangGraph to create an explicit stateful workflow.
- Maintains conversational state using SQLite checkpointing.

19 Limitations and Research Considerations

A research-style report should also state the limitations.

1. **No information system can guarantee that the user will “never miss” anything.** Search coverage, source availability, indexing delay, and model errors remain possible.
2. **Misinformation filtering is probabilistic.** The project reduces risk by preferring trusted/current sources and adding moderation, but it cannot guarantee truth.
3. **Priority is context-dependent.** A “Critical” item for one user may be irrelevant to another.
4. **FOMO is broader than social media.** The project should not claim that all FOMO is caused by social platforms.
5. **Information FOMO is a project framing.** It should not be presented as an officially recognized clinical subtype without supporting clinical literature.
6. **Current voice recognition is English-specific.** Multilingual voice input requires a multilingual ASR model or language-routing mechanism.
7. **Current agent count should be described accurately.** The code contains eight callable specialist tools and several graph-level nodes; “10+ agentic components” is more defensible than “10+ autonomous agents” for the present implementation.

20 Future Enhancements

Future versions can extend the system with:

- source credibility scoring;
- citation-aware answers and direct source verification;
- multilingual Whisper or automatic language detection;
- language-specific TTS voices;
- scheduled personalized digests;
- duplicate detection using semantic embeddings;

- temporal freshness scoring;
- user-specific priority models;
- explicit confidence scores;
- stronger fact verification using multiple independent sources;
- calendar/deadline integration;
- notification thresholds so users receive only high-priority alerts;
- analytics for measuring reduction in information volume and improvement in signal-to-noise ratio.

21 Evaluation Framework

A strong academic evaluation should measure the system rather than only demonstrate that it works.

Potential metrics include:

- **Relevance Precision:** proportion of delivered items judged relevant by the user.
- **Noise Reduction Rate:** proportion of retrieved items removed as irrelevant, duplicate, toxic, or low-value.
- **Opportunity Recall:** proportion of manually verified relevant opportunities successfully surfaced.
- **Priority Accuracy:** agreement between FOMO Priority classifications and human judgments.
- **Duplicate Suppression Rate:** percentage of repeated stories removed.
- **Latency:** time from query submission to response.
- **ASR Accuracy:** word error rate for supported speech conditions.
- **OCR Accuracy:** character/word accuracy on printed notices and documents.
- **Translation Quality:** human or automatic evaluation of multilingual output.
- **User Satisfaction:** perceived usefulness, clarity, and reduction in information overload.

I think it would be really helpful to try an experiment that looks at three ways of getting information. We can compare what happens when people can look at their media and news feeds without any limits when they have to search for information on their own and when they use Beyond FOMO.

22 Conclusion

Beyond FOMO is motivated by a genuine information-management paradox. Social-media consumption can expose users to social comparison, negativity, repetitive content, and engagement-driven noise. Completely leaving those platforms can reduce this exposure but may create concern about missing important information and opportunities.

The proposed middle path is not to maximize information consumption but to maximize **useful signal per unit of attention**. Through an Agentic AI architecture, the system searches for information, filters noise, evaluates relevance, prioritizes importance, identifies opportunities, moderates outputs, remembers preferences, and presents the result through text, voice, translation, and image-to-text interaction.

The central contribution of the project can therefore be summarized as:

Beyond FOMO does not attempt to keep the user connected to every feed. It attempts to keep the user connected to what actually matters.

References

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3. Project implementation: *fomo_agent.py*, Beyond FOMO LangGraph backend.
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